

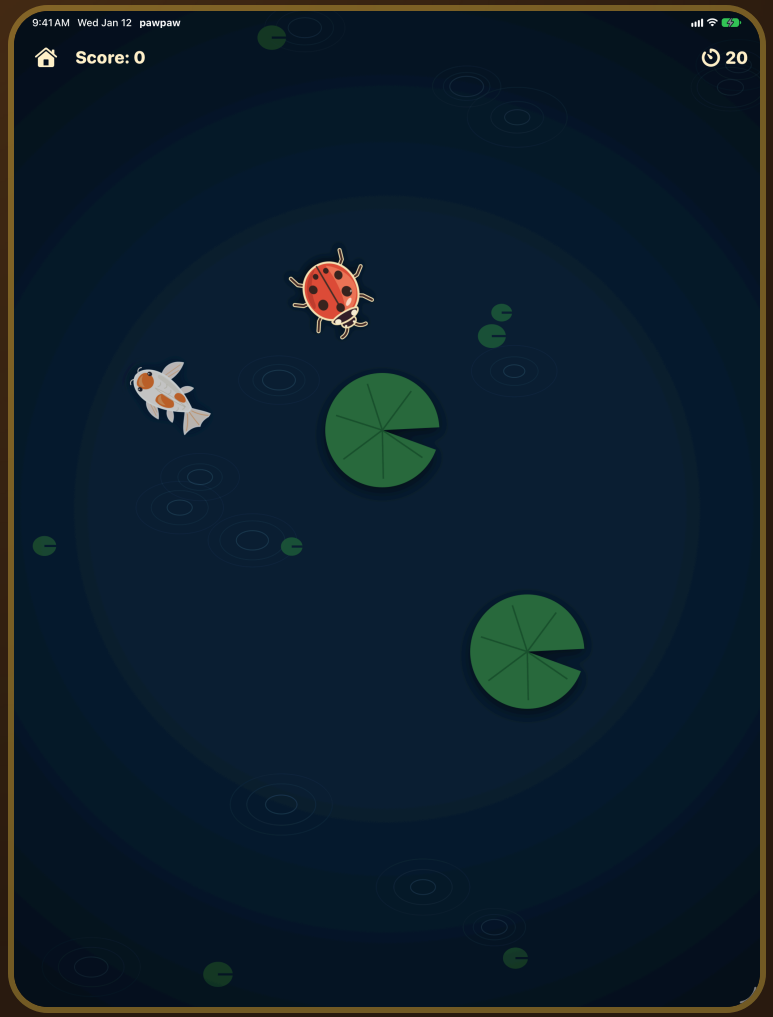
Paw Paw

An iPhone and iPad game made for cats

THE IDEA

One app, two players

- **The cat** plays: critters run across the screen, a paw catches them.
- **The owner** sets things up: round length, playground, cat profiles, stats.
- Neither of them can do the other's job, and that drove most of the design.



DESIGN THINKING

Who is each screen for?

SCREEN	WHO	SO IT…
Home, Stats, Profiles	**Owner**	Uses normal controls and follows the owner's text size and motion settings
First launch	**New owner**	Teaches by doing, not by reading
The game	**Cat**	Needs no reading, has big targets, and is hard to leave by accident
Between rounds	**Cat, then owner**	Lets the cat ask for more; the owner decides when to stop

DESIGN THINKING

Onboarding: learn by doing

The owner will hand the device to a cat, so they need to know how it works before that.



1

Catch

Try catching real critters on a small live field.



2

Countdown

Lay the device flat, 3-2-1, let the cat go.



3

Hold to leave

Practise the hold. A tap only shows a hint.



4

Your cat

Pick a face and a name. The title greets the cat as you type.

DESIGN THINKING

How does a cat say "again"?

The problem: a timed round ends, the cat is still hunting, and the owner may not be watching.

- One critter, the **lure**, keeps roaming around the results card.
- It hides under the card and peeks out from the edges.
- **Catching it starts the next round.**
- The owner picks: *Keep going*, *3 rounds* or *Stop*.

Ignores taps for 1 s, so a late pounce doesn't count

Wanders off after 45 s without a touch

Owners hold **Play Again** on the card

DESIGN THINKING

Moving like real prey

Dart and freeze

Each critter has its own style. Mice dart, crickets hop, feathers drift.

Twitch while still

Small twitches during a freeze keep the cat's eyes on it.

Hide and peek

Boxes and leaf piles hide critters, so the cat waits and pounces.

Heard first

Critters enter from off screen, with a sound from that side.

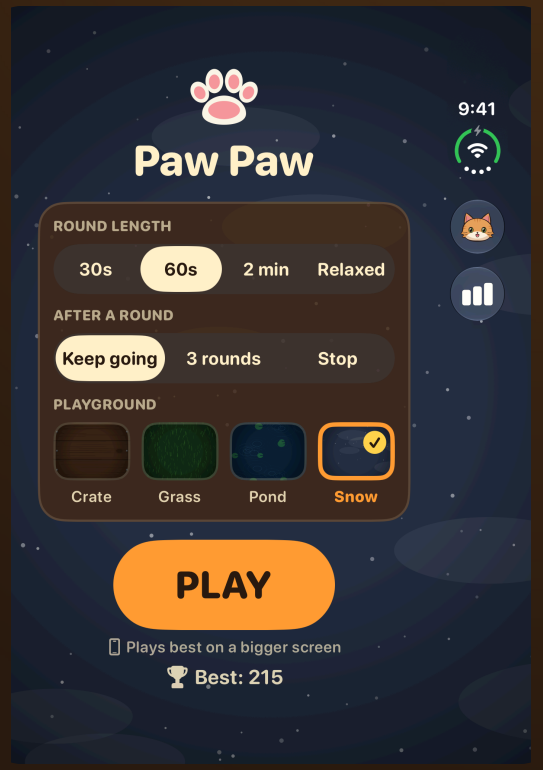


DESIGN THINKING

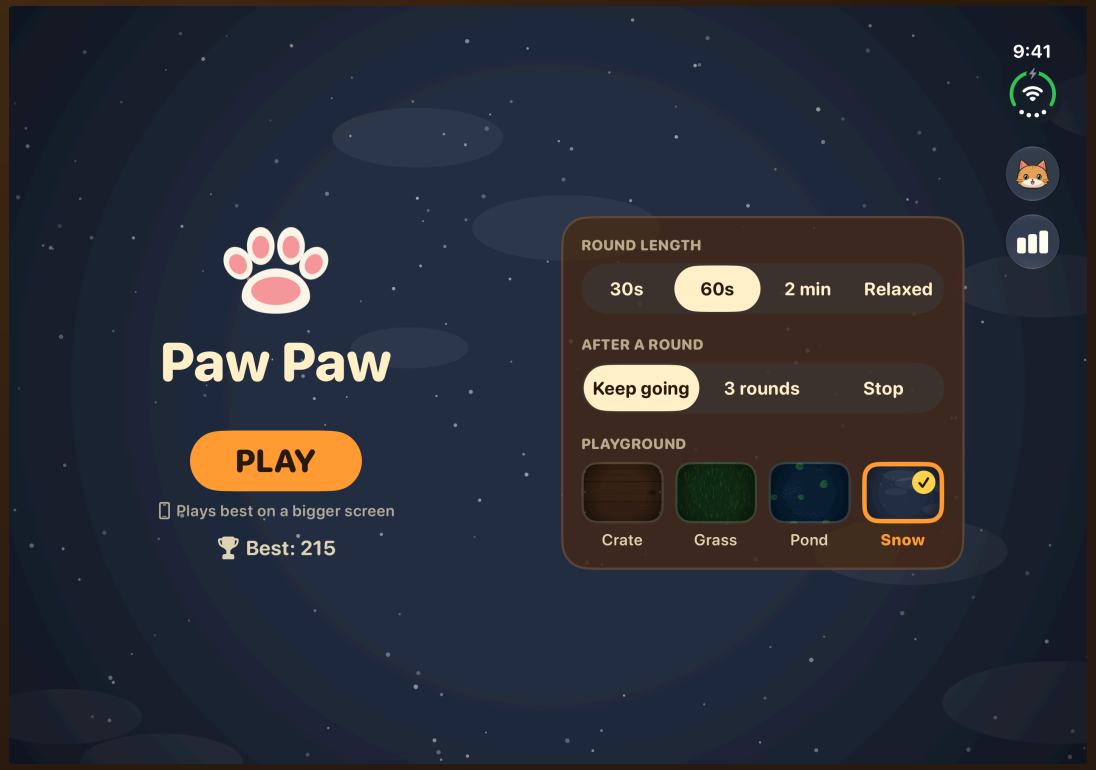
One layout per screen shape



iPhone
Menus portrait, game landscape, one critter at a time



iPhone Duo, folded
Same stacked layout as a phone



iPad & Duo, unfolded
Side by side in landscape, any orientation

ACCESSIBILITY, FROM CLASS

Whose settings apply?

The menus follow the **owner's** settings. The game is for the **cat**, so it only changes enough for the cat to keep playing and the owner to still get out.

VoiceOver

Every hold gesture is also a simple action: "Leave game", "Play again", "Delete cat", "Reset stats".

Direct touch

With VoiceOver on, the play field still takes taps right away, so the cat can play.

Dynamic Type

Menu text grows up to 200% and screens scroll. The game's score and timer stop at 125%.

Reduce Motion

Calms the menus and intro, but not the cat's game.

LIVE DEMO

Let's play

- First launch: onboarding and cat setup
- Home: round length, after a round, playground
- A round: critters, hiding spots, hold to leave
- Between rounds: the lure
- Stats and cat profiles



BUILDING IT

Why not just UIKit?

- UIKit redraws when something **changes**: great for buttons, lists and forms.
- A game changes **every frame**: up to 120 times a second, with critters moving, hiding and being caught.
- So I kept UIKit for everything the owner touches, and used **SpriteKit**, Apple's built-in 2D game framework, for the playfield.

UIKit: score, timer, Home button

SpriteKit: critters, hiding spots, effects

UIKit: drawn background (pond, grass, snow)

SPRITEKIT

If you know UIKit, you know SpriteKit

UIKIT	SPRITEKIT	WHAT IT DOES
`UIView`	`SKNode`	Something on screen, inside a tree of parents and children
`UIImageView`	`SKSpriteNode`	Shows a picture, like a mouse or a beetle
`UIView.animate`	`SKAction`	"Move here over 1 second", without writing the frames yourself
`touchesBegan`	`touchesBegan`	The same method, because nodes are UIKit responders too

`SKView` is a `UIView`, so the whole game sits inside a normal view controller.

TAKEAWAYS

What I learned

- Start with *who is holding the device*, screen by screen.
- When a user can't read, the design has to show, not tell.
- Lay out for the space you have, not the device name.
- UIKit for the app, SpriteKit for the action, and they fit together.



Thanks!

Questions?